



Analytical Solutions For Flexural Design Of Hybrid Reinforced Concrete

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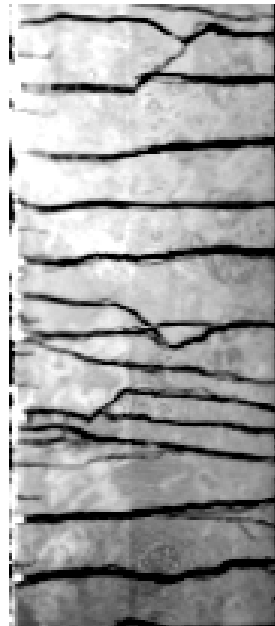
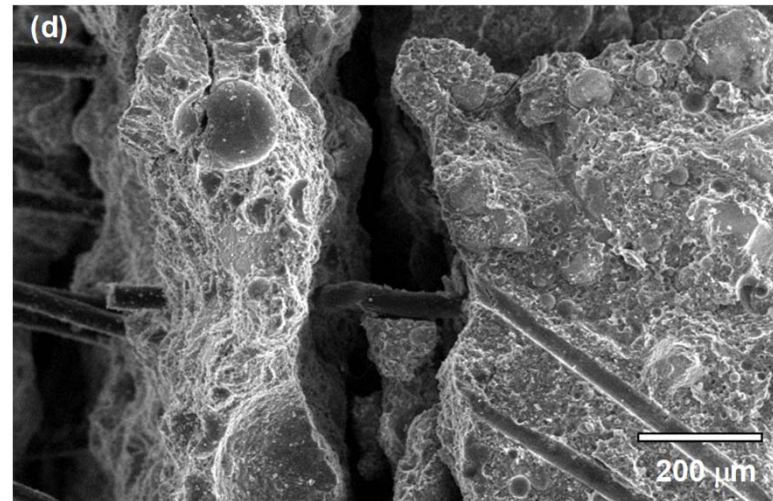
TRB, 2020

Opportunities in use of FRC in Serviceability Based Structural Design Applications

- The phenomenal growth of fiber reinforced concrete market is a key motivator for addressing sustainability-based design
- Economy, labor, time, materials characteristics and performance
- Recent developments have played a significant role in developing documents to showcase the performance of FRC materials
- Design opportunities:
 - Ductility, durability, crack width, stiffness, cracked section modulus.
 - Shear
 - A hybrid approach of combining reinforcement and fibers is the key to addressing sustainability
 - Minimum reinforcement requirements.

Materials characteristics

- Ductility
- Toughening
- Improved tensile strength
- Increase level of energy absorption
- Fatigue life, impact/explosive loading
- Seismic resistance
- Steel work, labor, construction time.
- Corrosion damage
- Long-term repair and maintenance.



**Design at the microstructure level will have a profound impact
at the macro-structural level**



Applications for UHPC Bridge Applications

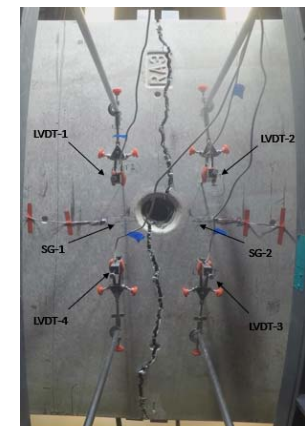


ADOT SPR 745 Project

Precast Tunnel lining hybrid design



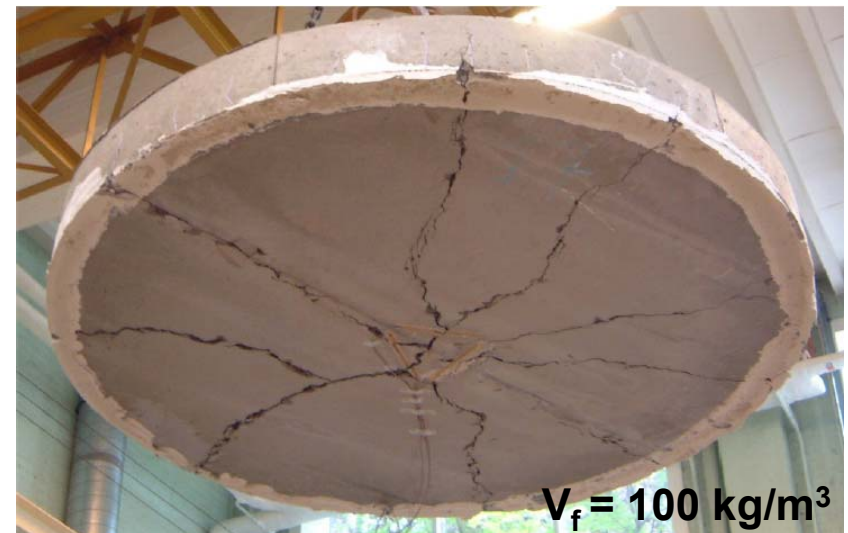
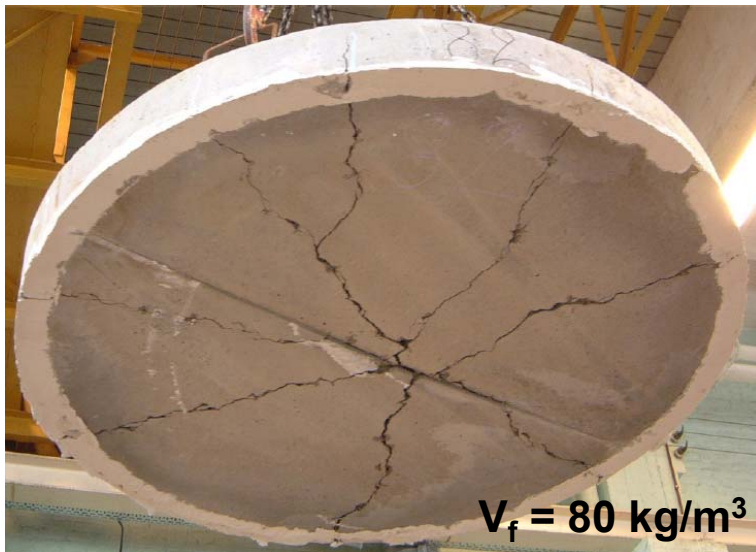
Large Frame Testing setup at ASU



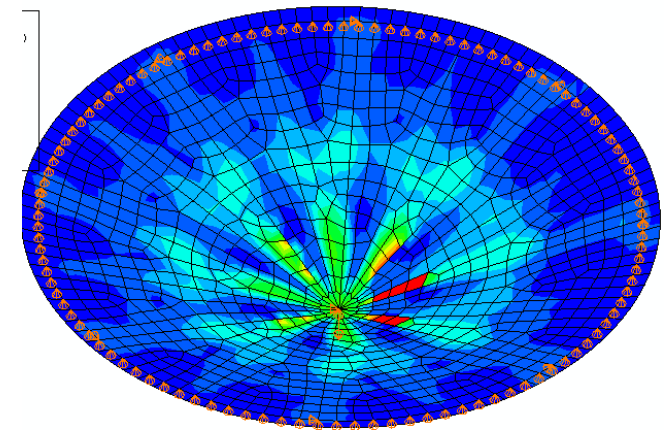
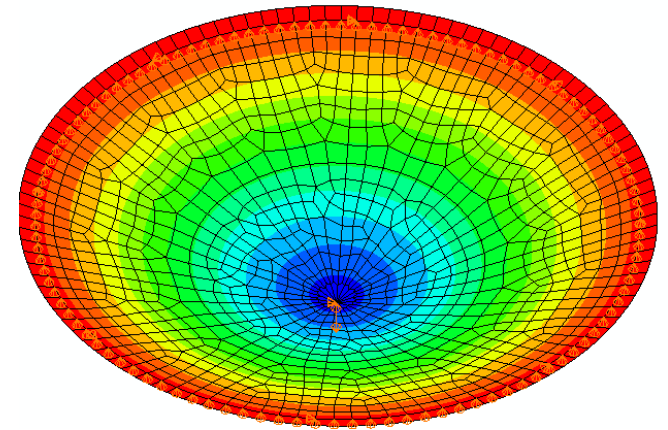
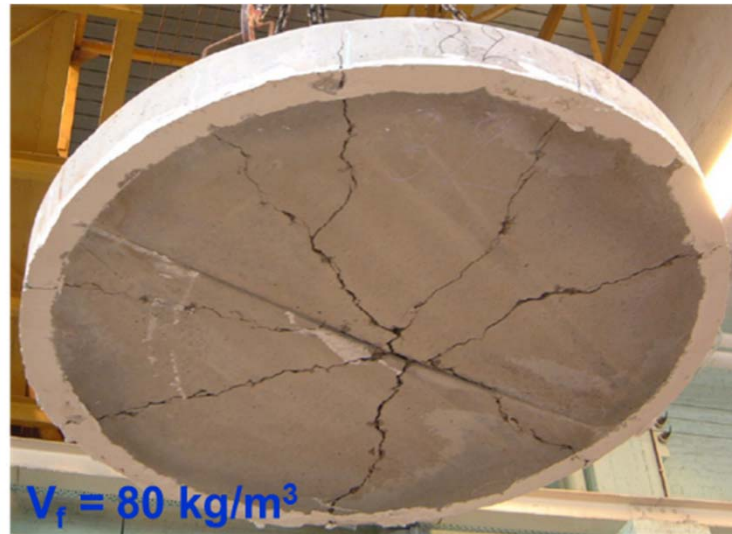
The location of measuring device

Round panel tests for evaluation of SFRC

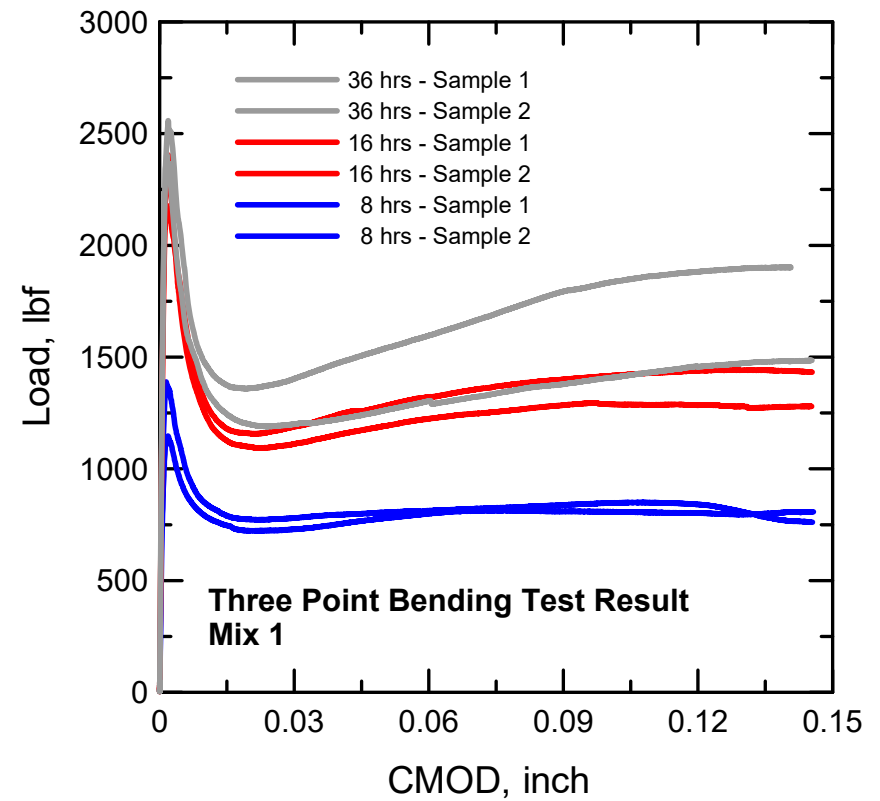
- Test setup
 - displacement controlled, continuous support, center point load
- Dimensions
 - D, t= 1500, 150 mm



Round Panel Continuous Support Specimens

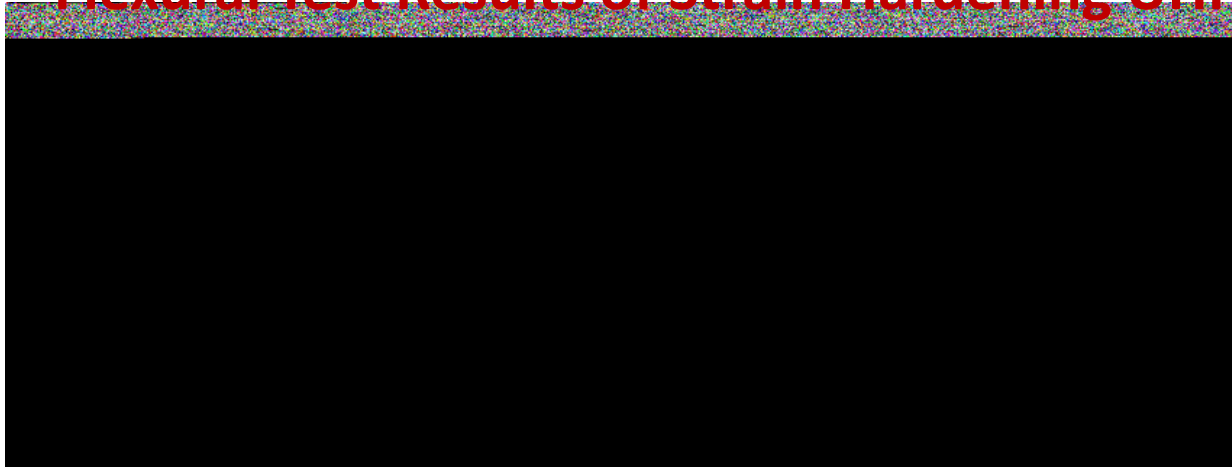


Residual strength Effect in Strain Softening beams

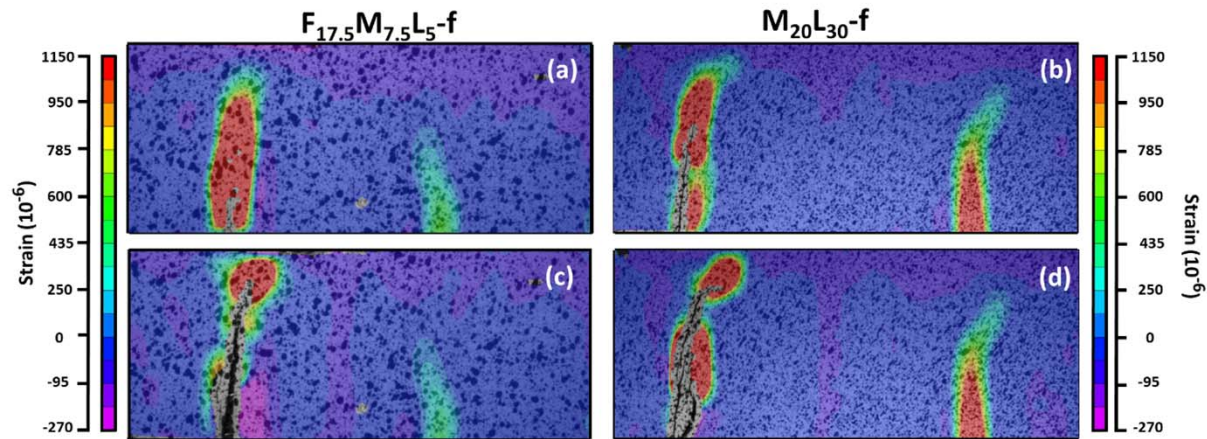


10-12 lbs/yd³ of macro fibers

Flexural Test Results of Strain Hardening UHPC Beams

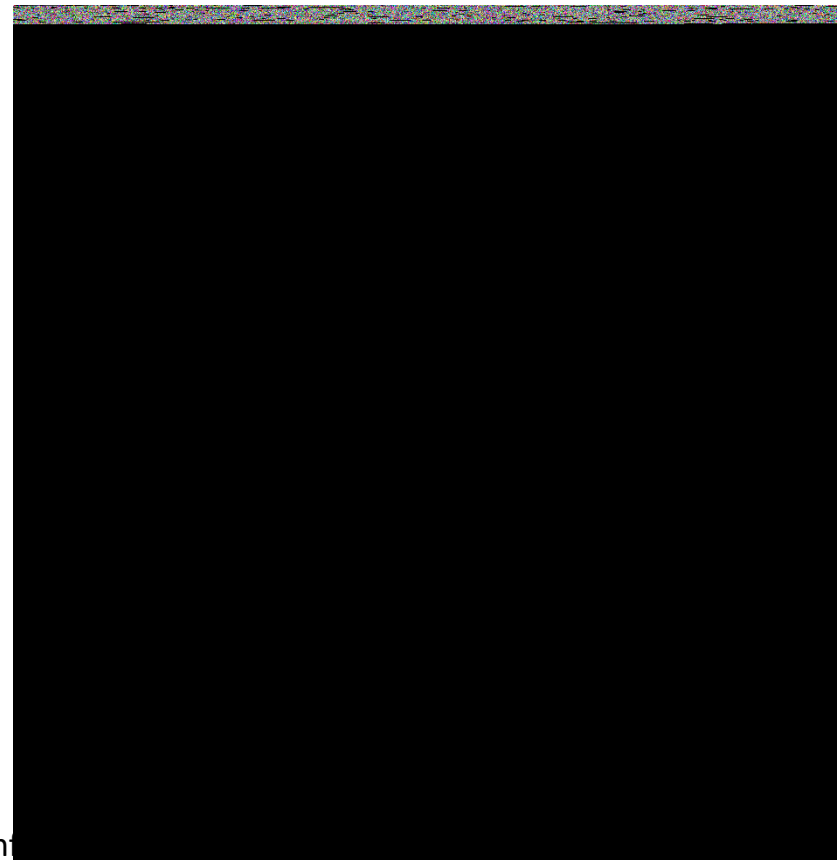
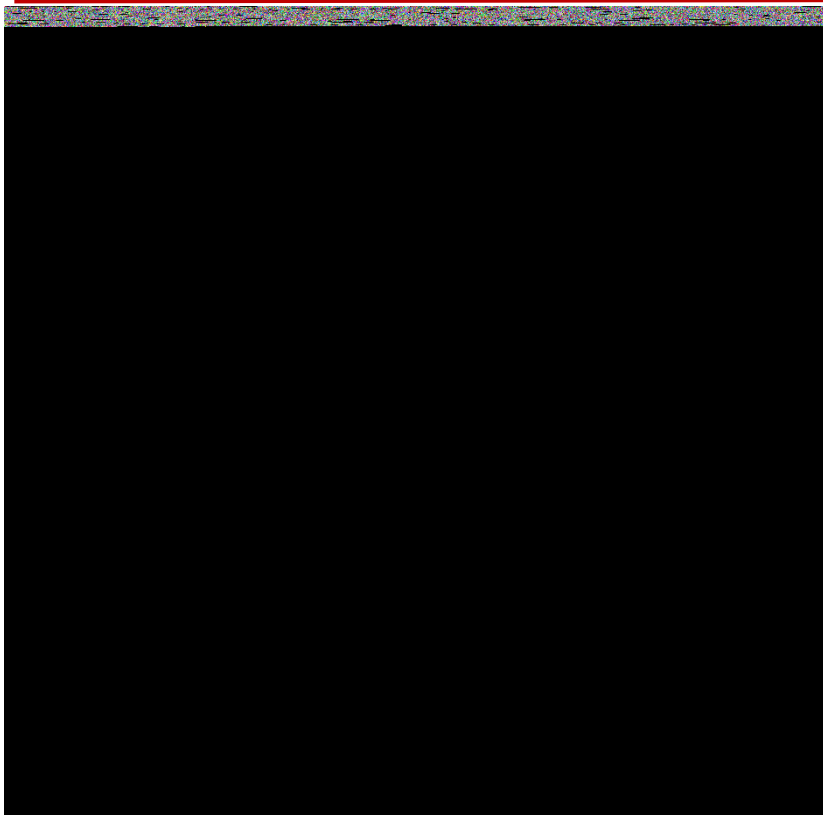


strain distribution in UHPC beams at peak load just prior to failure



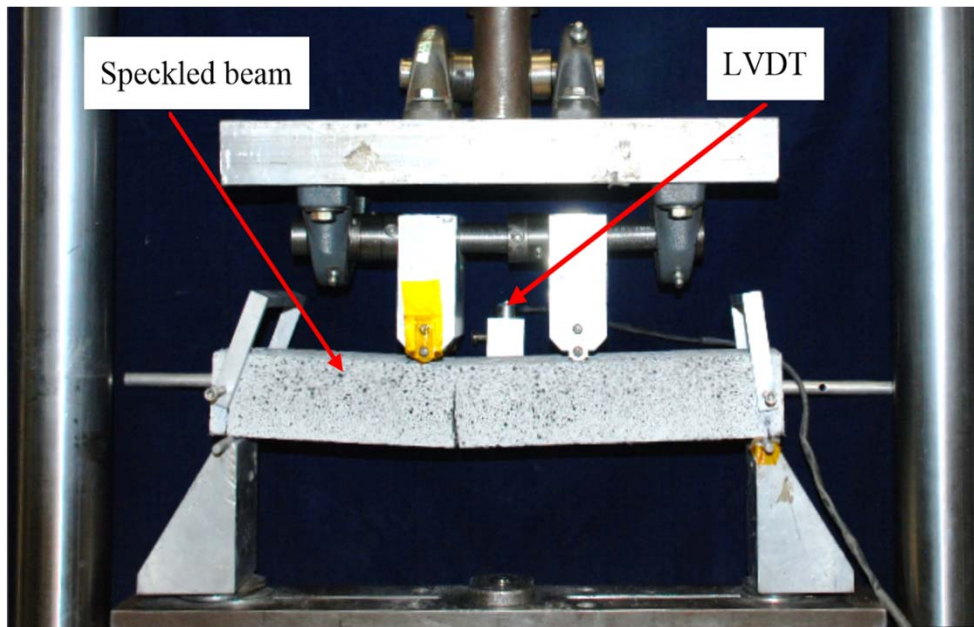
Strain distribution in FRC beams with 1% steel fiber, post peak stress (a,b) 90% and (c,d) 50% post-peak stress.

Plain UHPC and Effect of Fiber Reinforcement on Strain Hardening Behavior

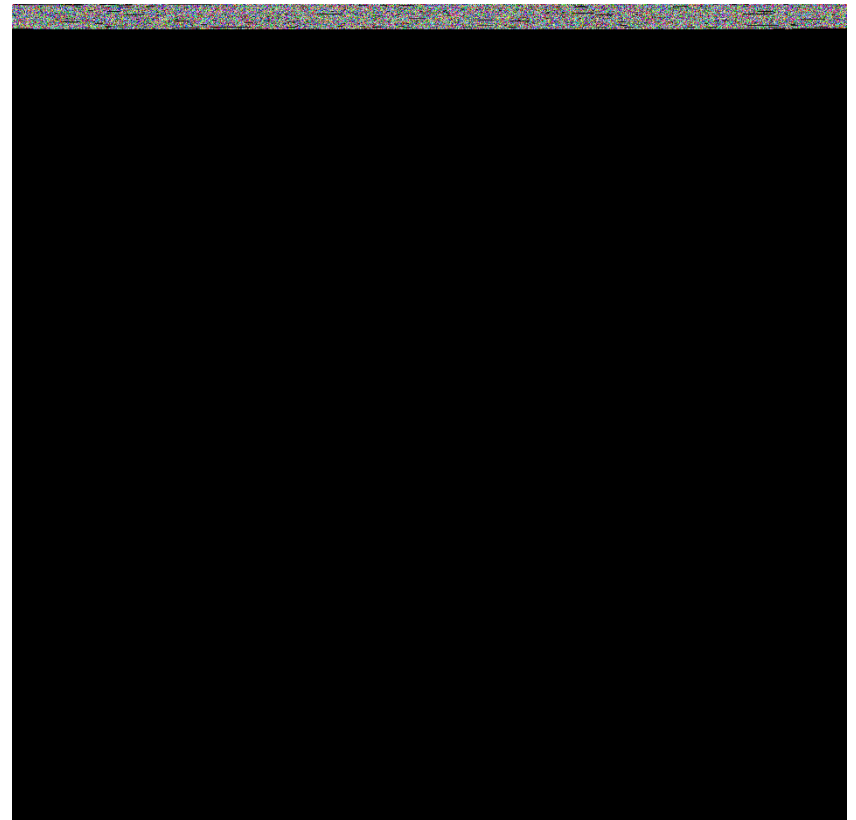


- No post-peak response for the brittle unreinforced UHPC
- Considerable non-linear response after the occurrence of the first crack with 1% fibers

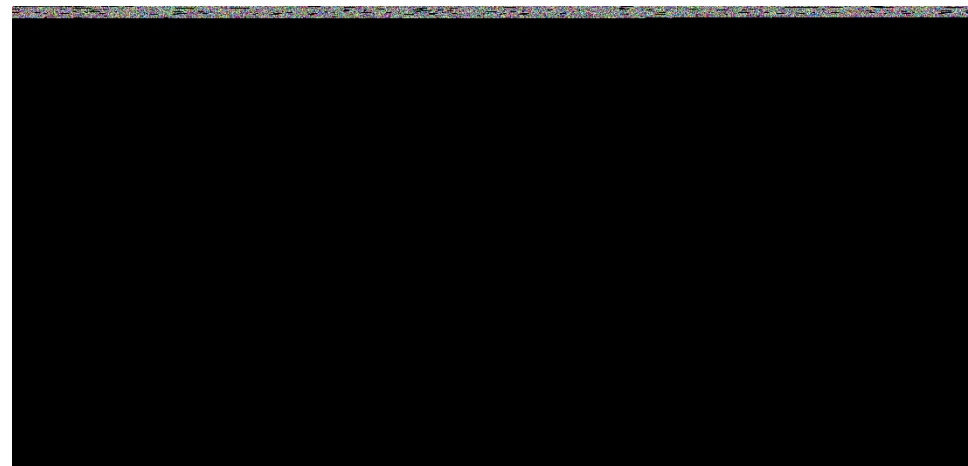
Four-point bending test of Strain Hardening materials



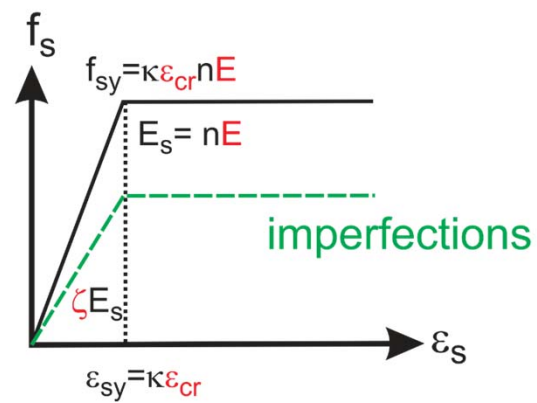
Set up for flexural Four-point bending test



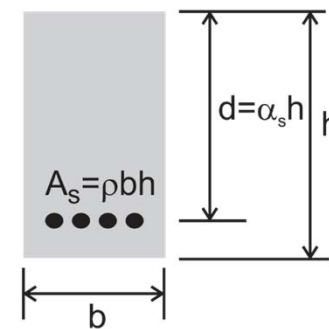
Material Models



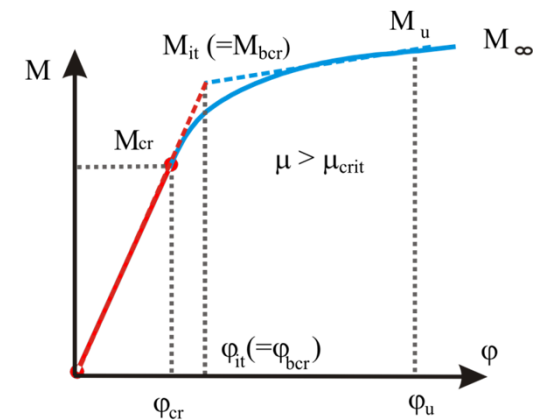
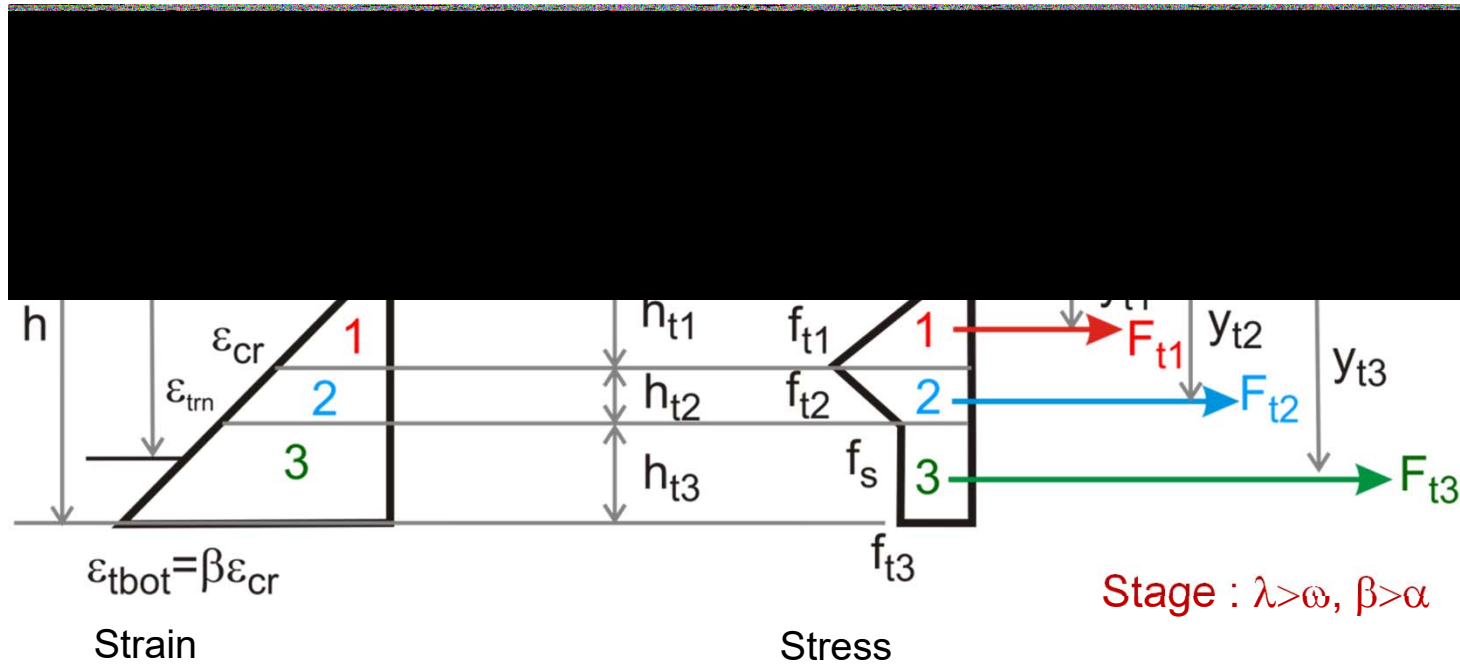
Steel rebar



Cross section



Derivation of Moment-Curvature Relationship



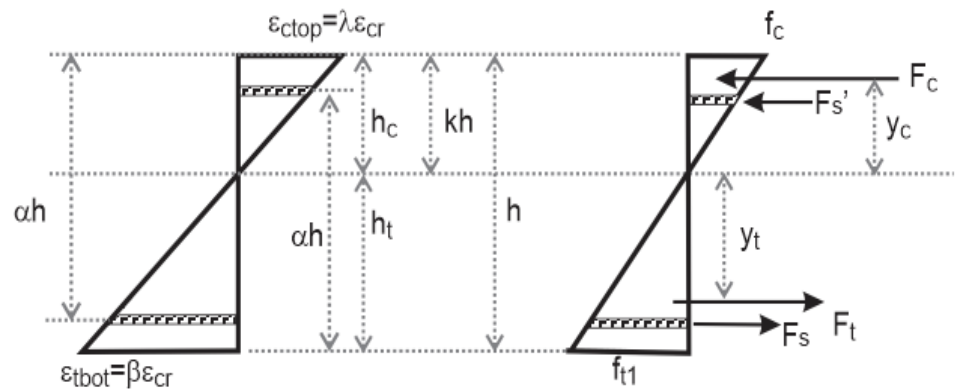
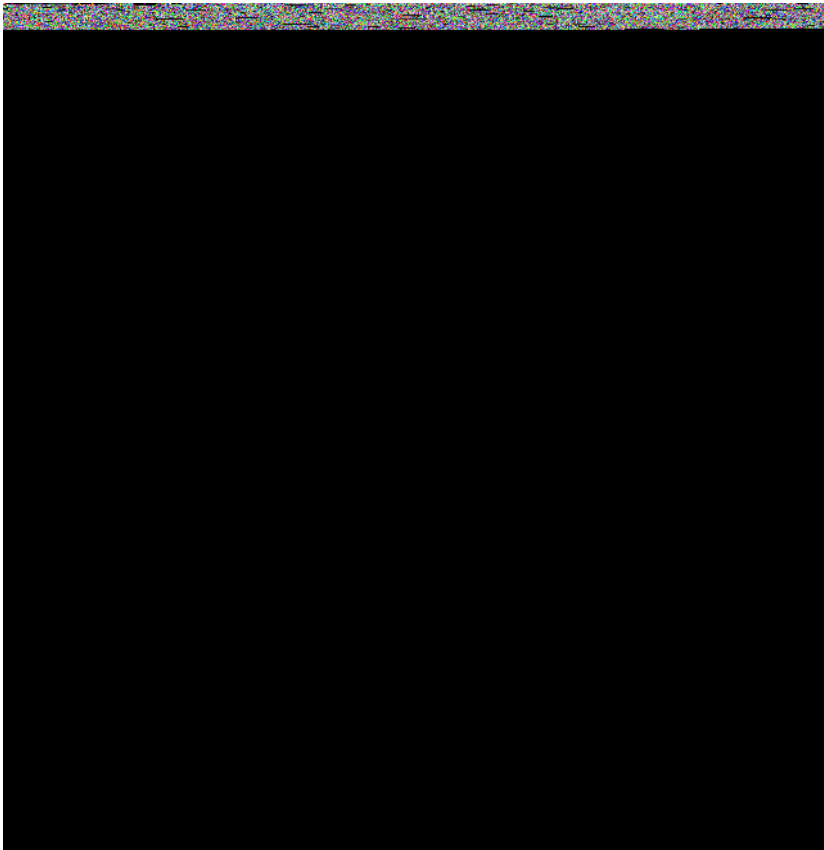
Incrementally
impose
 $0 \leq \epsilon_t \leq \epsilon_{tu}$

Strain Distribution
Stress Distribution

$\Sigma F = 0$,
determine k

Moment: $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$
Curvature: $\phi = \epsilon_c / kh$

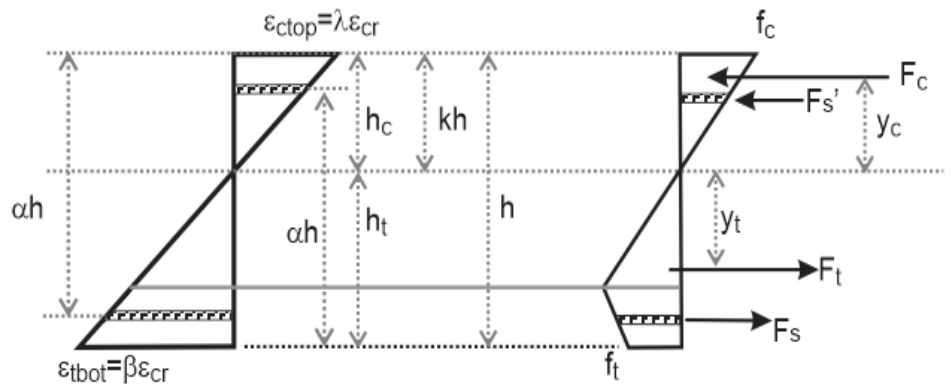
Strain and stress distribution at any stages



Zone1 :

- compression and tension are in elastic zone
- Steel is not yield

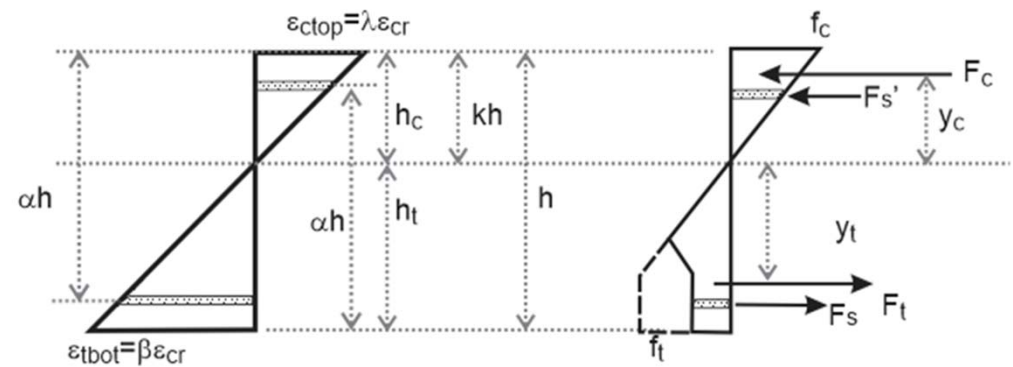
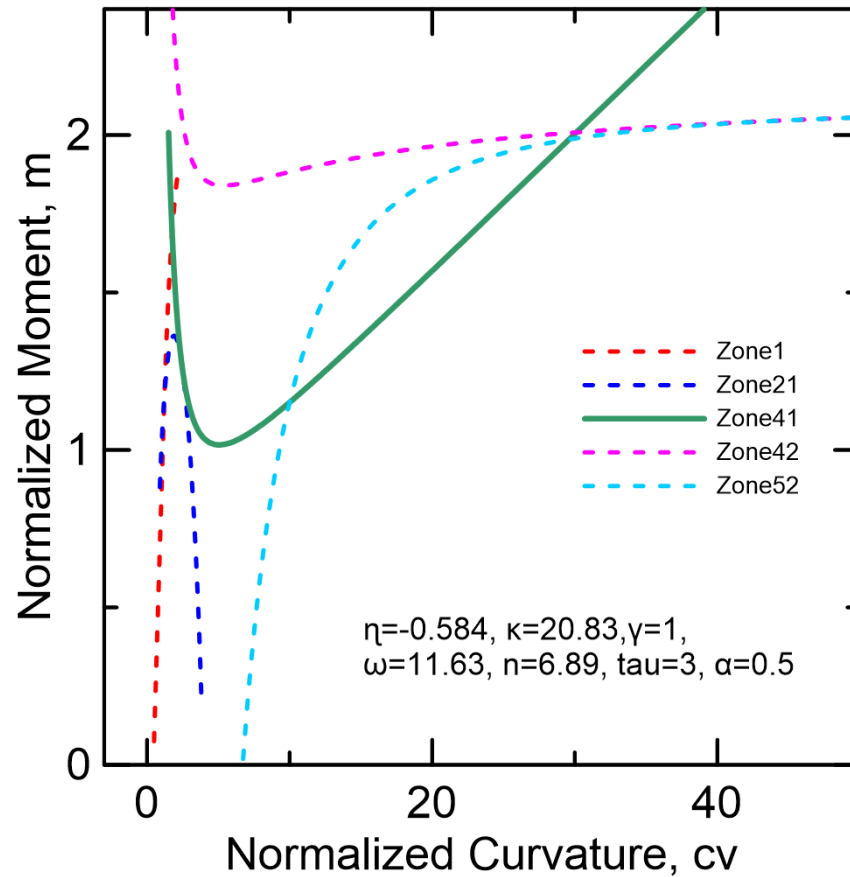
Zone2.1



Zone21

- compression is in elastic zone
- tension is second slope zone
- Steel has not yielded

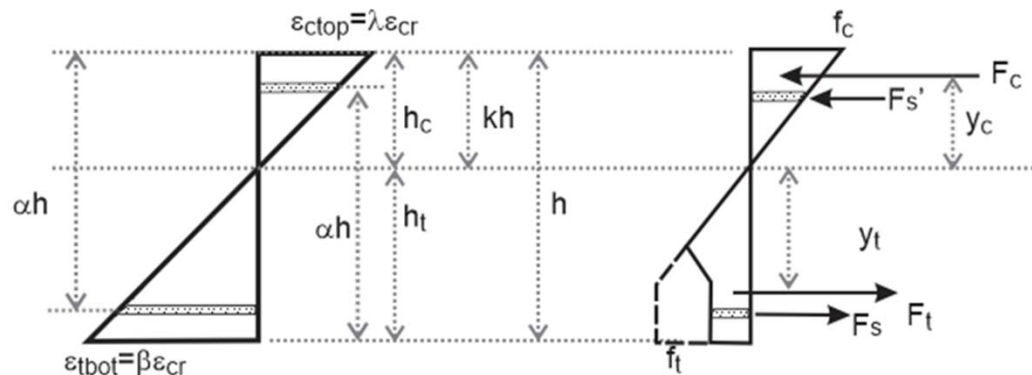
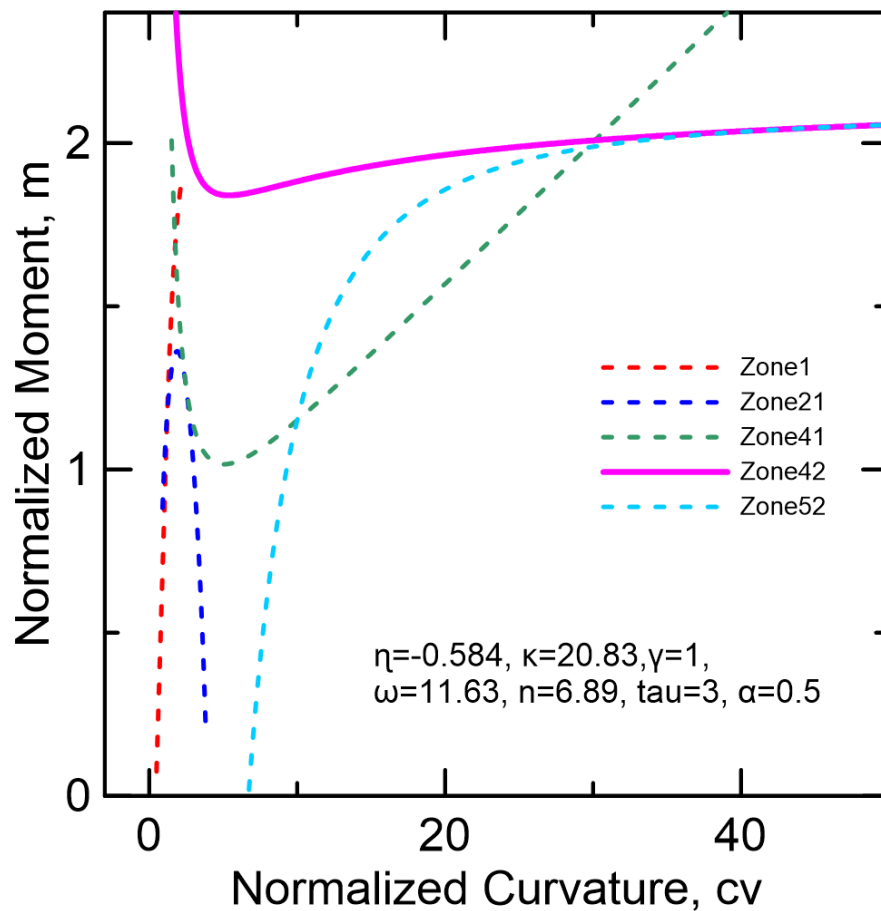
Zone4.1



Stage41 :

- compression is in elastic zone
- tension is in residual range
- Steel has not yielded

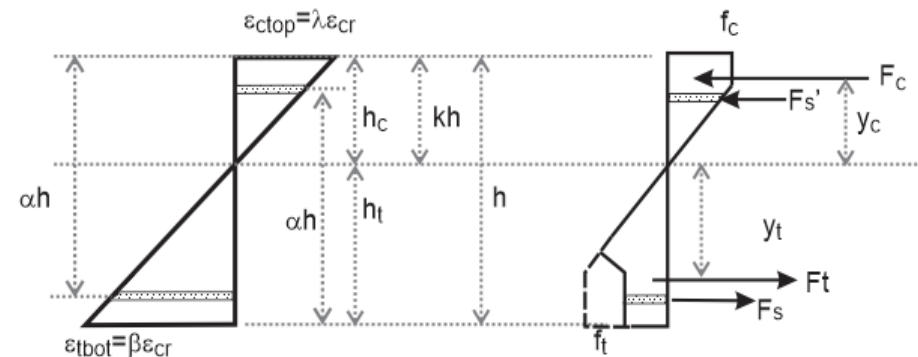
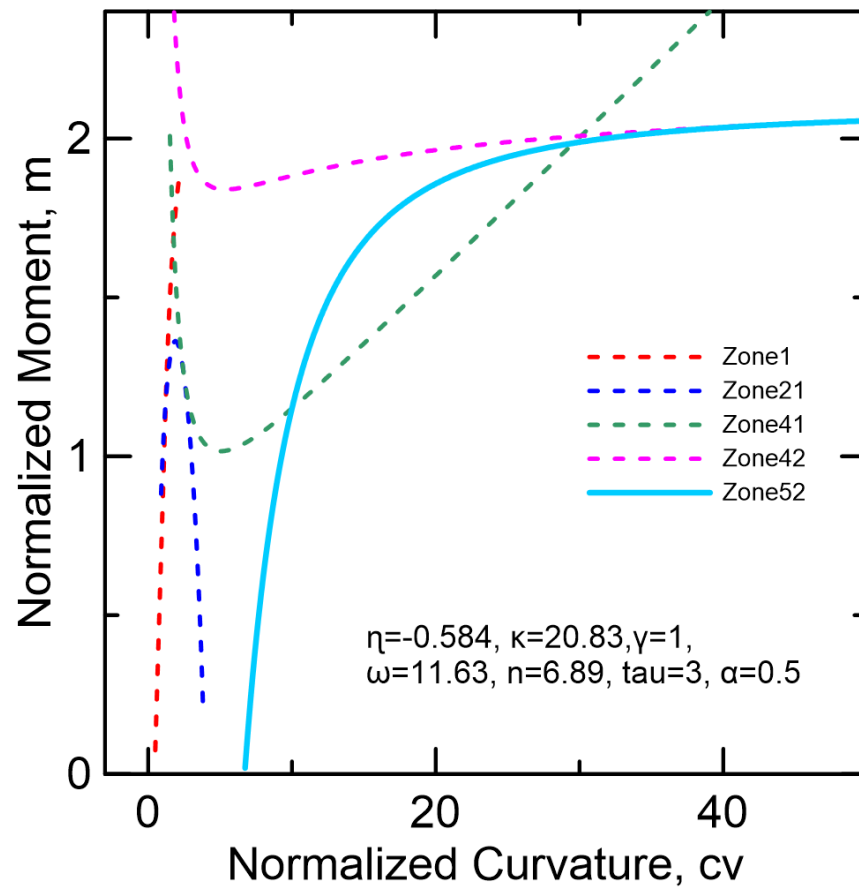
Zone4.2



Stage42 :

- compression is in elastic zone
- tension is in residual range
- Steel has yield

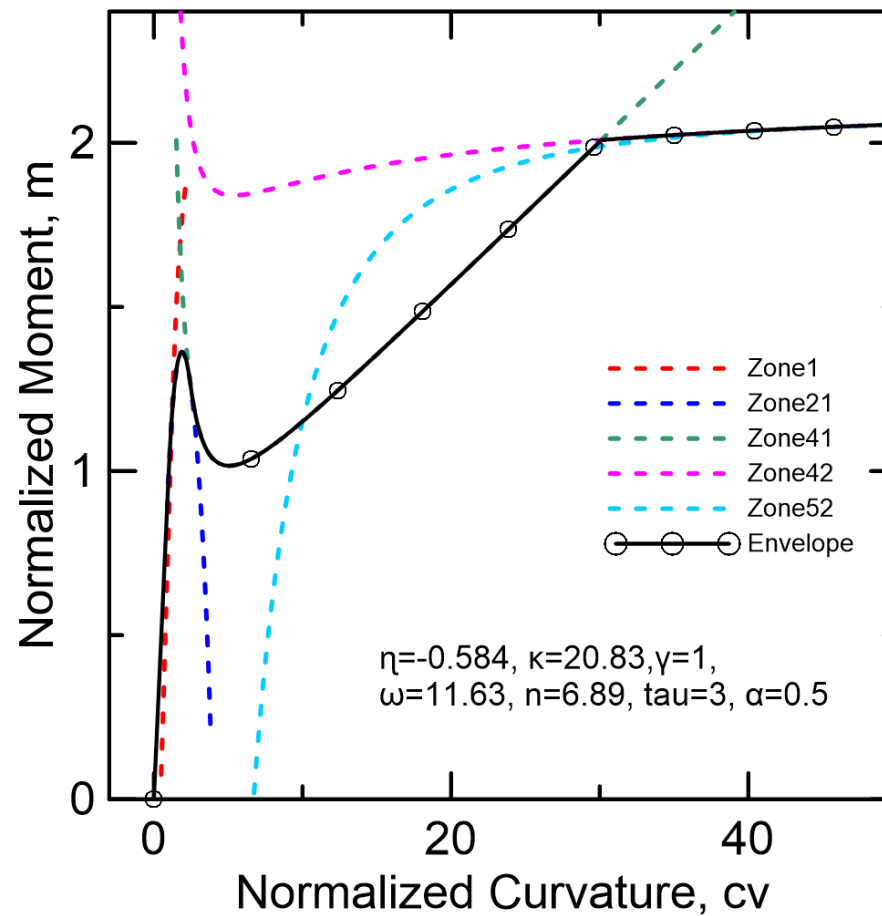
Zone5.2



Stage52 :

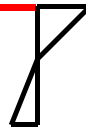
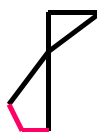
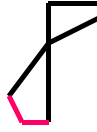
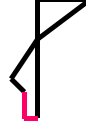
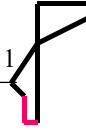
- compression is in plastic zone
- tension is in residual range
- Steel has yielded

Envelope curve



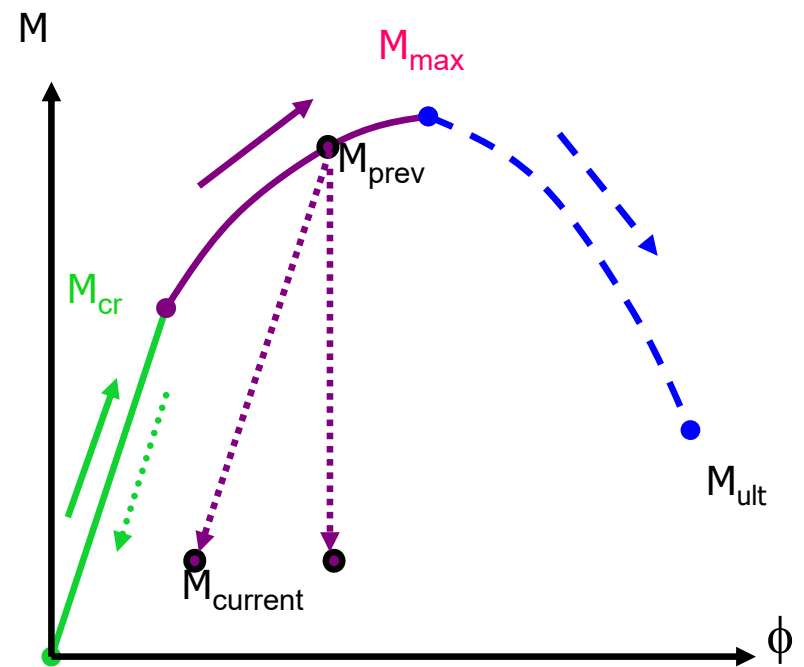
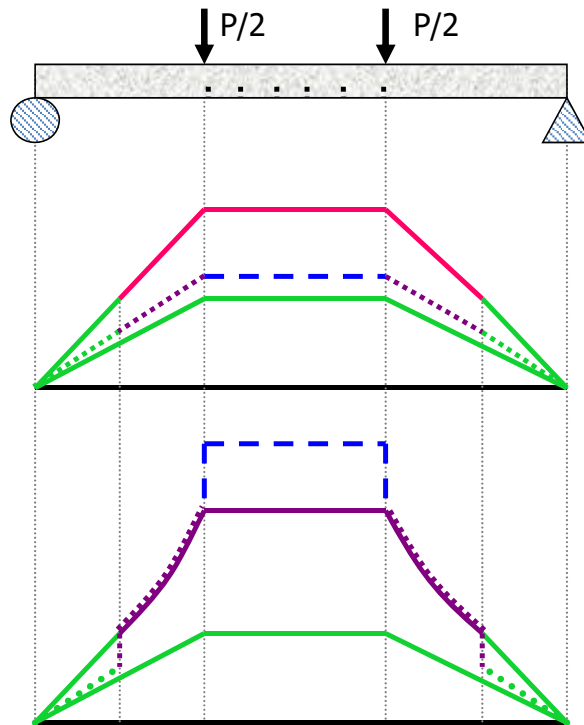
Moment Curvature Diagram for -strain hardening –softening materials

$$\phi' = \phi / \phi_{cr} \quad \phi'_i = \frac{\beta}{2(1 - k_i)}$$

Stage	Range	k	$M' = M / M_{cr}$	
1	$(0 < \beta \leq 1)$ and $(\lambda \leq \omega)$	$k_1 = \begin{cases} \frac{-1 + \sqrt{\gamma}}{-1 + \gamma} & \text{for } \gamma > 1 \\ \frac{1}{2} & \text{for } \gamma = 1 \end{cases}$	$M'_1 = \frac{2\beta[(\gamma - 1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1 - k_1}$	
2.1	$(1 < \beta \leq \alpha)$ and $(\lambda \leq \omega)$	$k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$ $D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$	$M'_{21} = \frac{(2\beta\gamma + C_{21})k_{21}^3 - 3C_{21}k_{21}^2 + 3C_{21}k_{21} - C_{21}}{1 - k_{21}}$ $C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$	
2.2	$(1 < \beta \leq \alpha)$ and $(\omega < \lambda \leq \lambda_{cu})$	$k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$ $D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$	$M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$ $C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$	
3.1	$(\beta > \alpha)$ and $(\lambda \leq \omega)$	$k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$ $D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$	$M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$ $C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$	
3.2	$(\beta > \alpha)$ and $(\omega < \lambda \leq \lambda_{cu})$	$k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$ $D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$	$M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$ $C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$	

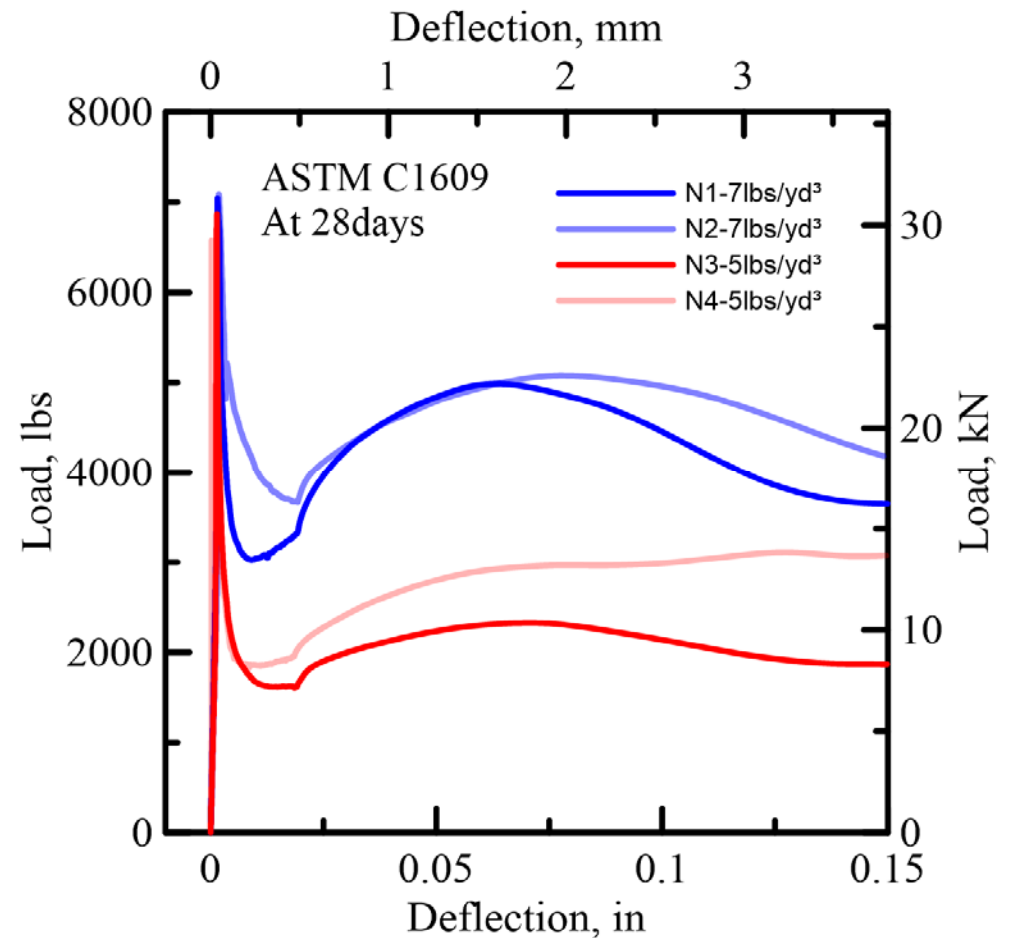
Load Deflection Computation

- Moment-curvature diagrams are calculated for individual zones of response
- Equilibrium is used to obtain support reactions and moment distribution
- The rotation and deflection distributions are obtained by the slope deflection principal

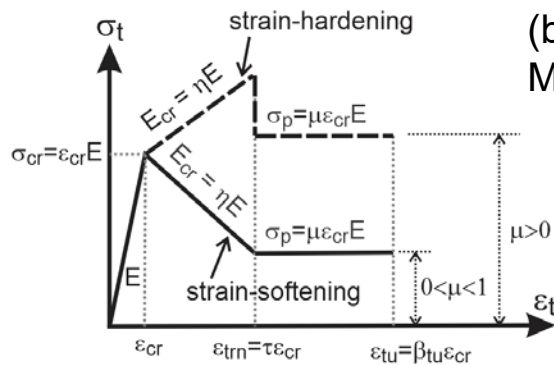
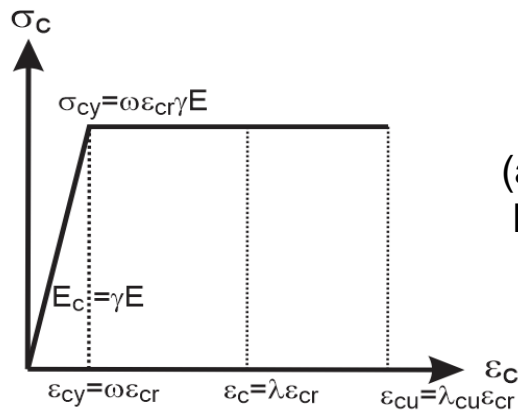


Experimental result

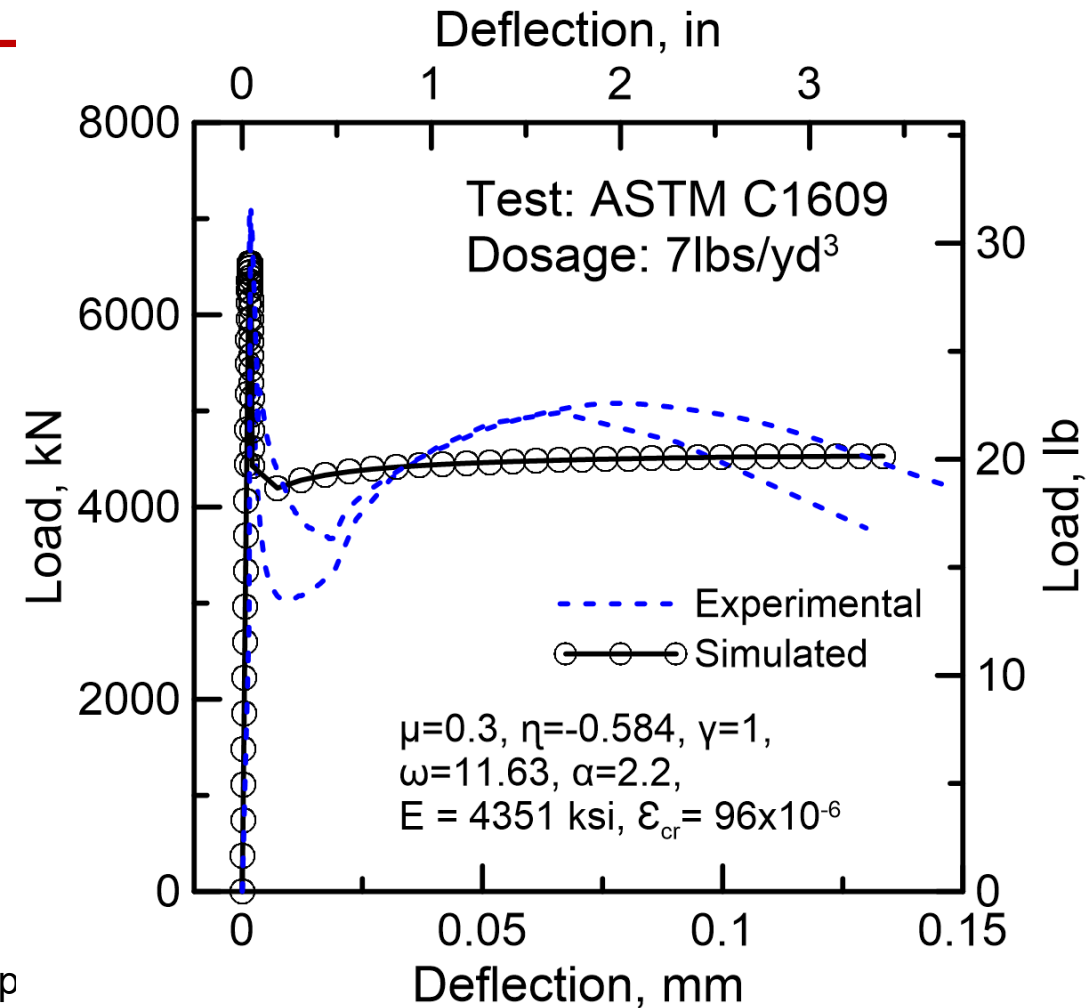
ASTM C1609 experiment from 5 and 7 lbs per cubic yard of macrosynthetic fibers



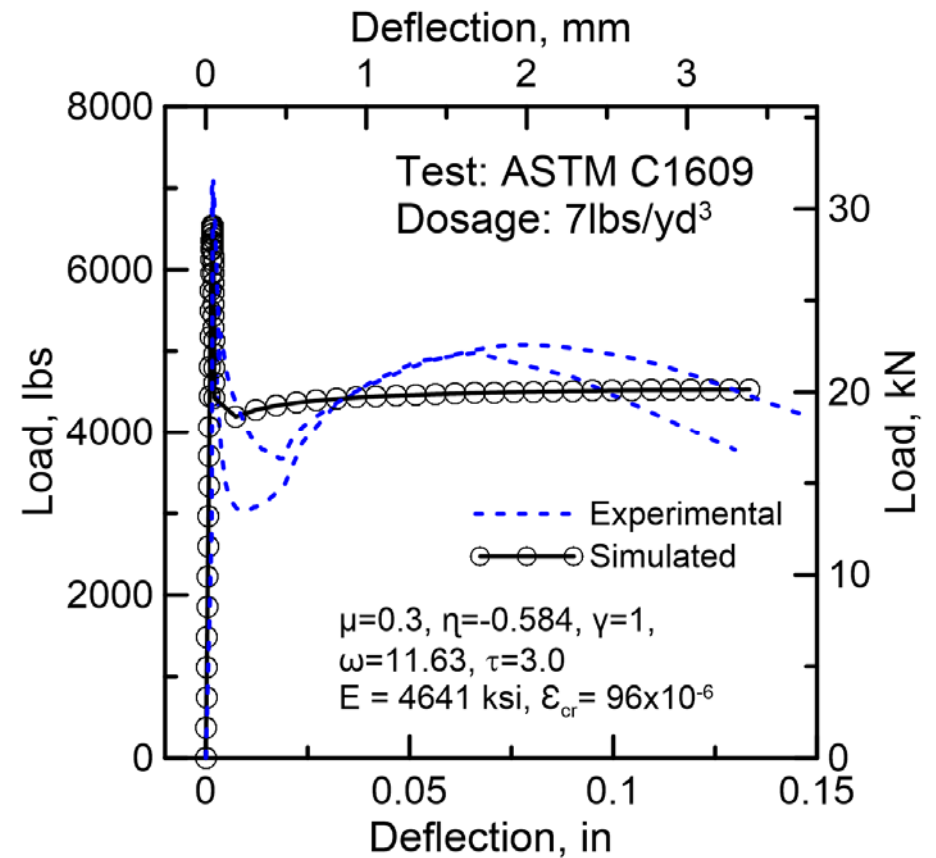
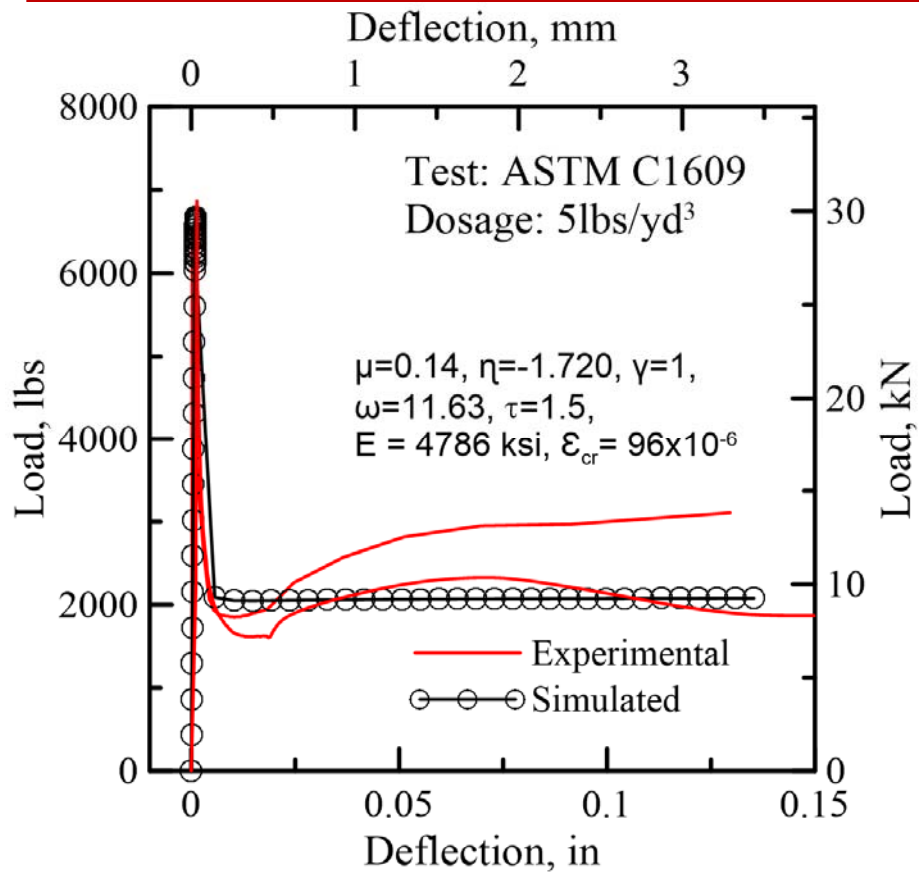
Develop a parameterized approach



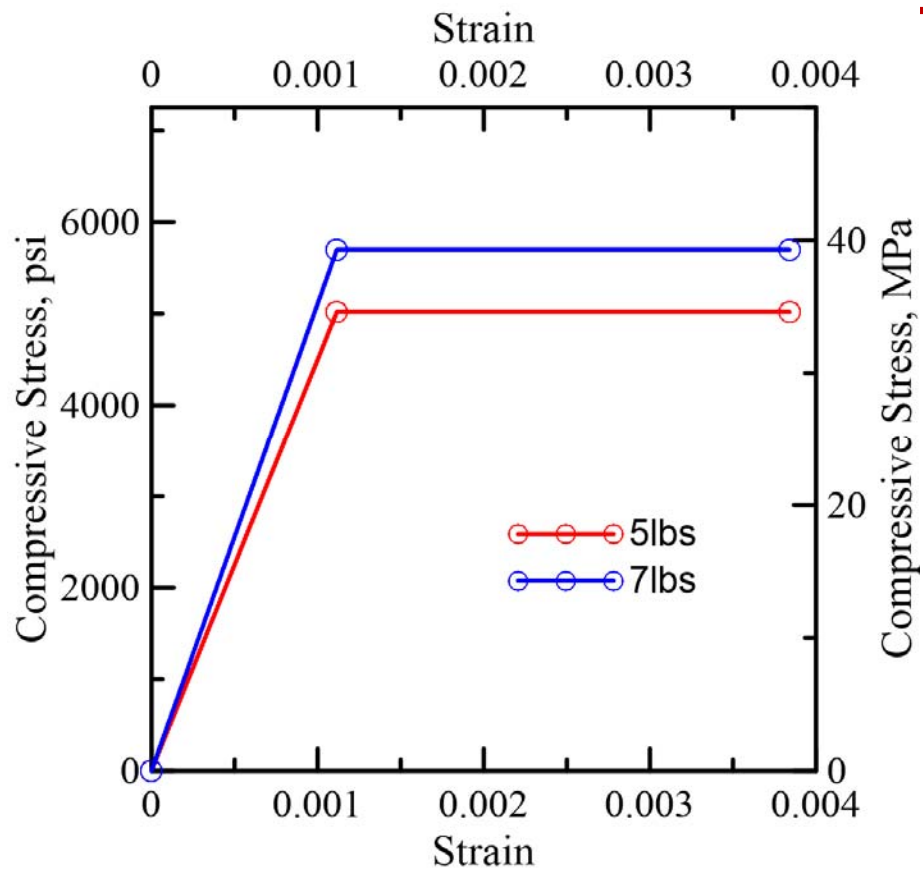
- Simulate L-D using parametric in tension and comp



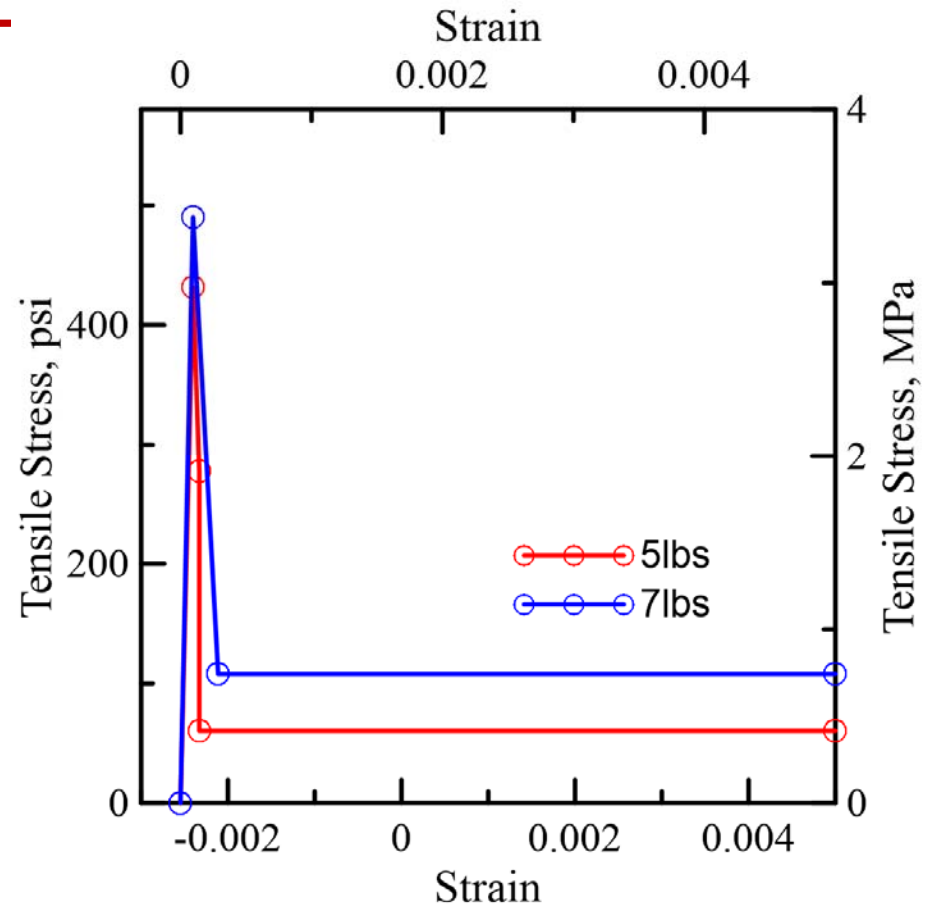
Experimental result



Backcalculated Constitutive Models



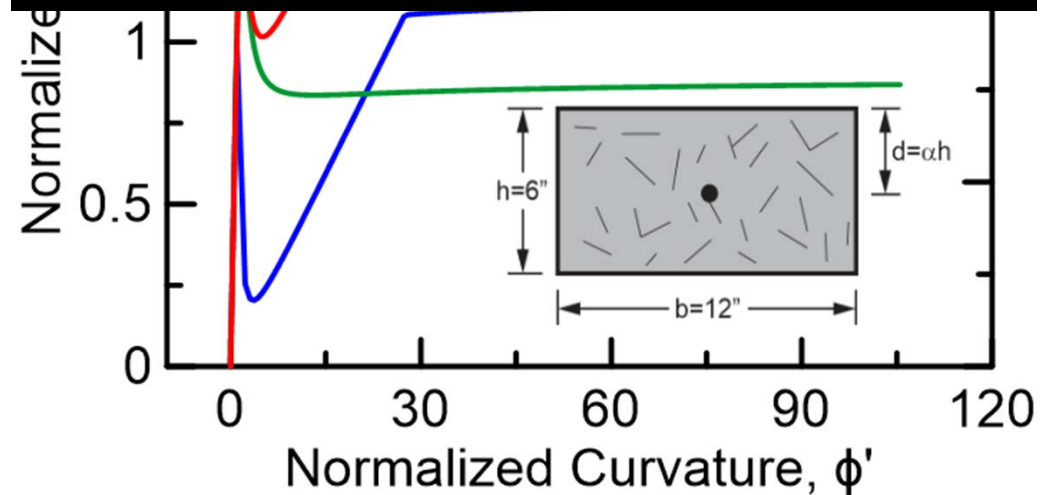
(a) Compressive Stress Model



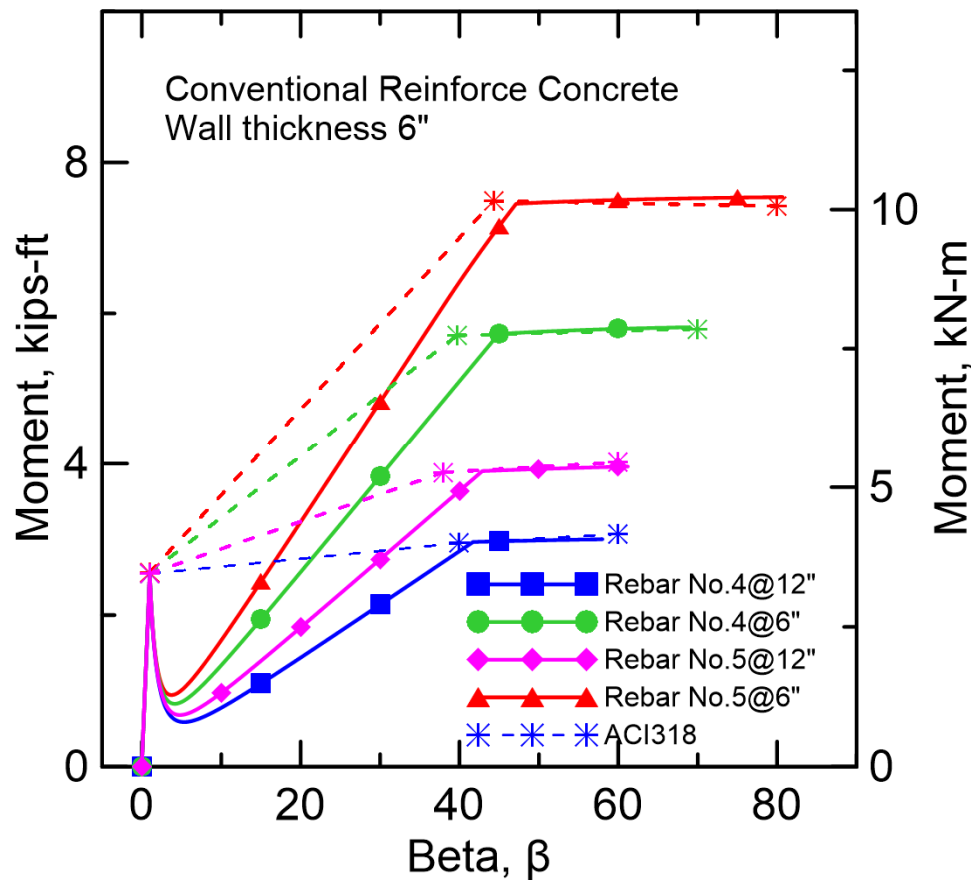
(b) Tensile Stress Model

Three main contributions of the Model Representing the Synergy of fiber and rebar combination

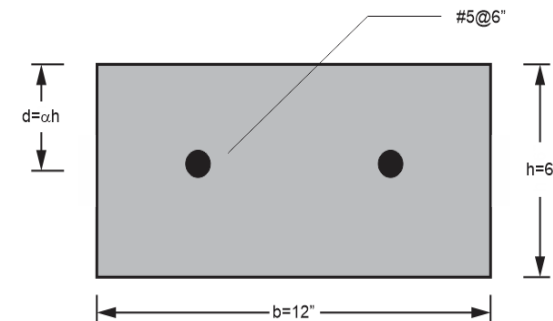
1. Simulated Moment-Curvature Response comparing HRC with plain FRC and conventional rebar
2. Additive contribution of fiber and steel based on compatibility of strain analysis
3. Capability to use a serviceability-based limit strain or curvature analysis



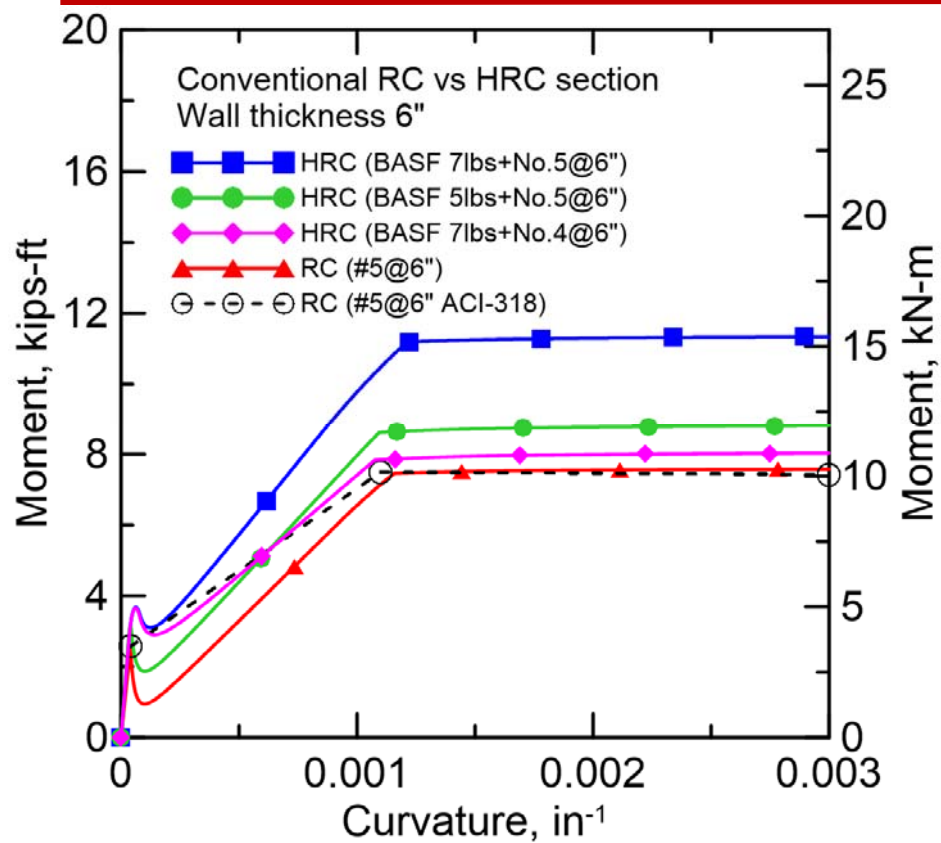
Modeling comparing conventional RC with ASU closed-form solution and ACI318-14 code



- Conventional RC (No Fiber) moment capacity of wall $t = 6"$ generated by ASU closed-form solution comparing with ACI318-14 code
- β is normalized tensile strain.
- To design serviceability limit state, We adjust beta values to satisfy moment capacity



Effect of fiber and rebar



(a) Moment capacity of wall section 8 in. computed with ASU closed-form solution, using fiber 7lbs/yd³



(b) Water tank segment with rebar

Conventional panels replacement

Original Design	Alternative Design	Reduction in reinforcement (%)	Reduction in concrete (%)	Fiber Dosage, pcy
#4@12 t= 8"	#4@12", t=6", 5PCY	0	-25%	5
	#4@12", t=6", 7PCY	0	-25%	7
	No rebar- t=8" +7PCY	-100%	-	7
#4@6 t=8"	#5@12 , t=8" ,5PCY	-23%	-	5
	#4@12 , t=8",7PCY	-50%	-	7
#5@12 t= 8"	#4@12 , t= 8", 5PCY	-35%	-	5
	#5@12, t=6", 7PCY	-	-25%	7
	#4@12, t=6", 5PCY	-35%	-25%	5
	#4@12, t=8",7PCY	-35%	-	7

(a) Summary table for replacing conventional panels with HRC/FRC panels

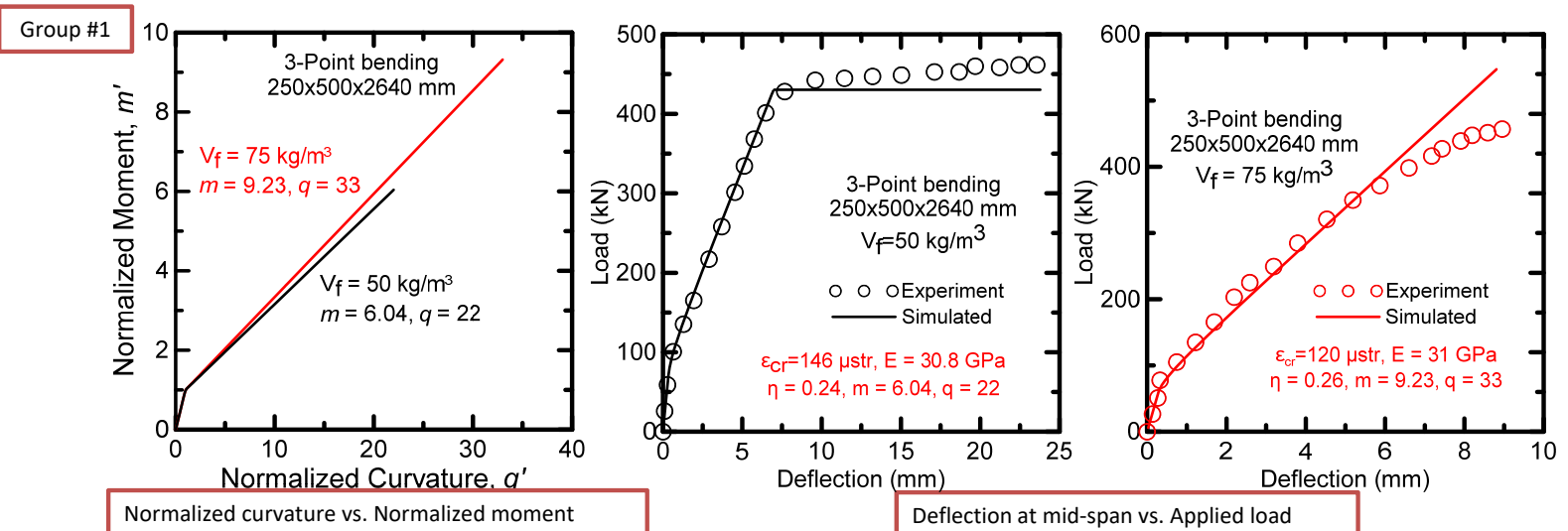
Experimental Verification – RC with Steel fibers

- Three-point bending test
- Fiber content: 50 and 75 kg/m³

Beams	Group #1	Group #2	Group #3
Height (mm)	500	1000	1500
Span(mm)	2640	5640	8640
Width(mm)	250	250	250
Reinforcement Longitudinal bars	8-φ14	8-φ20	8-φ24

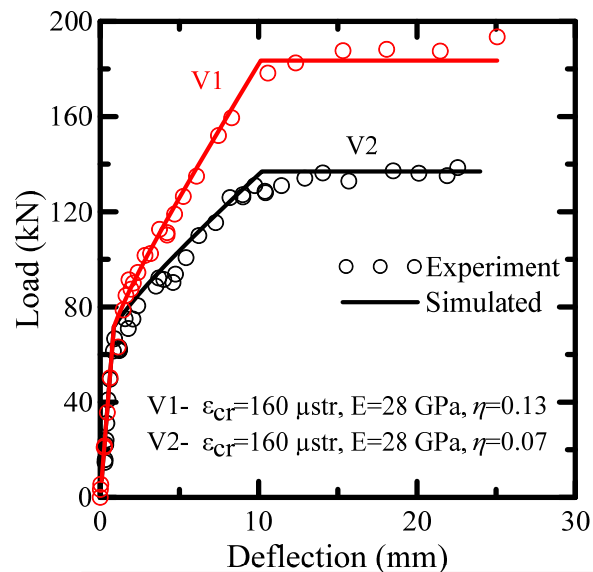
Simulated Parameters

Beam Type	Fiber Content kg/m ³	ϵ_{cr}	E	η	m	q	$EI_g, 10^{13}$	$EI_{cr}, 10^{13}$
		10 ⁻⁶	GPa					
Group #1	50	146	30.8	0.24	6.04	22	8.02	1.93
	75	120	31	0.26	9.23	33	8.07	2.10
Group #2	50	120	30	0.22	5.18	20	62.5	13.8
	75	120	31	0.28	6.88	22	64.6	18.1
Group #3	50	120	29	0.24	6.04	22	204	48.9
	75	120	30	0.28	7.16	23	211	59.1



Experimental Verification – RC with Steel fibers

- Four-point bending test
- 200(b) × 600(d) × 4000 (L) mm
- Steel fibers
- Longitudinal and shear reinforcement
- Measure deflections by five LVDTs
- $x = 750, 1500, 2000, 2500, 3250$ mm



Deflection at mid-span vs. Applied load



Details of beams

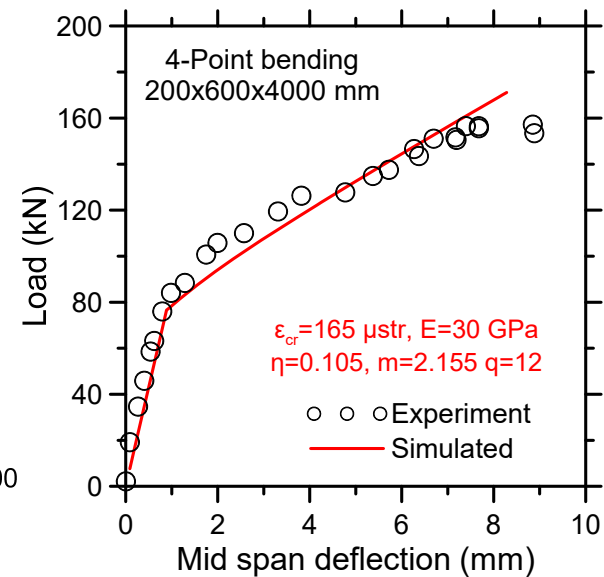
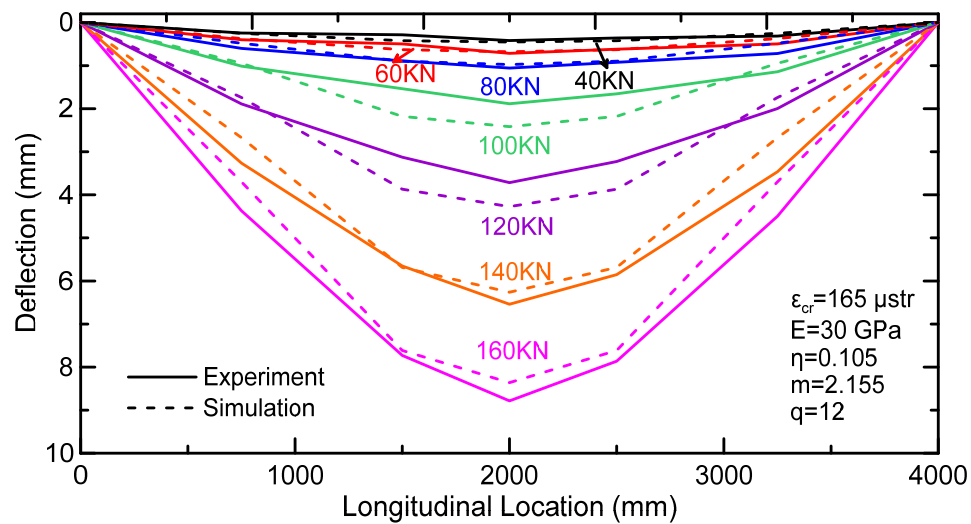
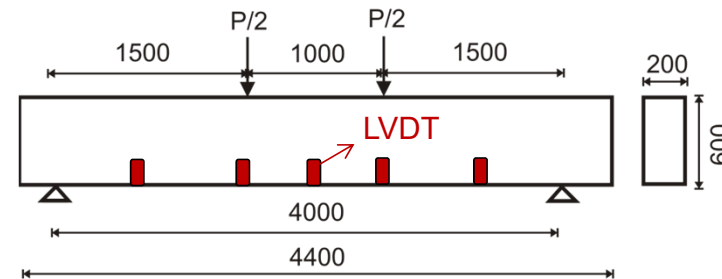
Beam ID	Fiber volume fraction (%)	Longitudinal reinforcement			Shear reinforcement	
		Ratio (%)	Tension	Top bar	Ratio (%)	Rebars
V1	0	0.262	4 ϕ 10	2 ϕ 6.3	0.104	4 ϕ 6.3
V2	0	0.262	4 ϕ 10	2 ϕ 6.3	0	0

Simulated Parameters

ID	m	q	$EI_g, 10^{14}$	$EI_{cr}, 10^{14}$
V1	2.56	100	1.01	0.131
V2	1.91	92	1.01	0.071

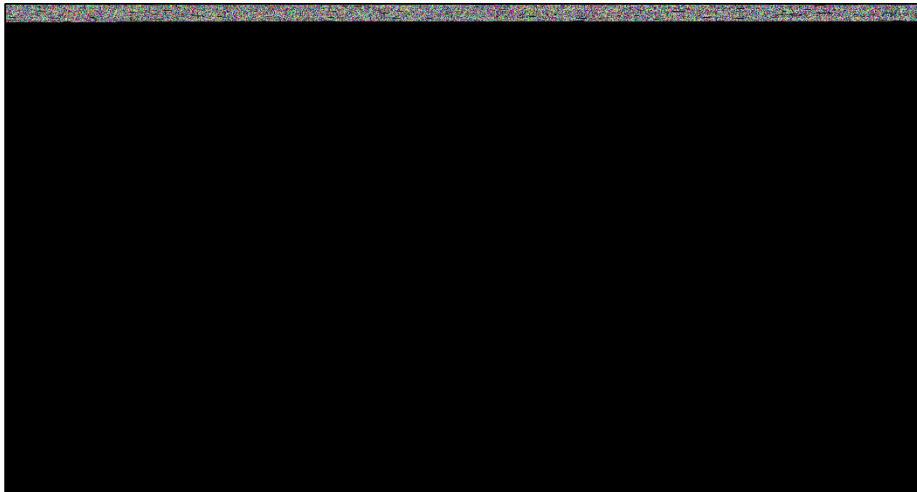
Experimental Verification– RC with Steel fibers

- 4m span
- Fiber volume fraction : 1.25%
- 2 ϕ 10 (Tension bars), 2 ϕ 6.3(Comp. bars)
- Pre-cracked stiffness $EI_g = 1.08 \times 10^{14}$
- Post-cracked stiffness $EI_{cr} = 1.13 \times 10^{13}$

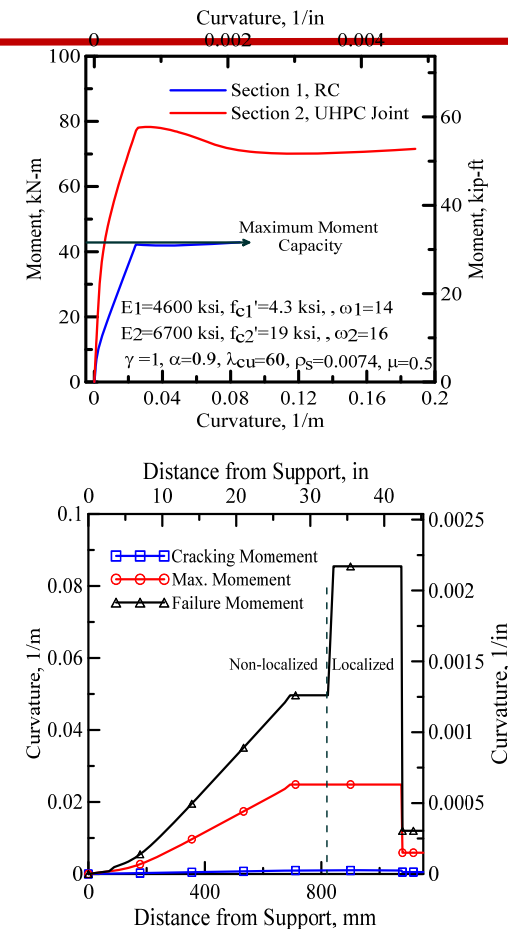


Evaluation of UHPC Connection Pours

- Two sections modelled, regular RC and UHPC
- Each section with its own moment-curvature response
- The moment capacity is dominated by the capacity of the weaker section (RC section)
- These curves show that making stronger joints will not help to improve the capacity of the member or slab



Haber et al., executed an experimental study on six different commercially-available materials (UHPC type A to F)



Full Scale Tunnel Lining Testing

Full-scale flexure and compression tests conducted on fiber-reinforced precast tunnel lining segments at ASU,



(a)



(b)

Casting Process of FRC precast segments



(c)

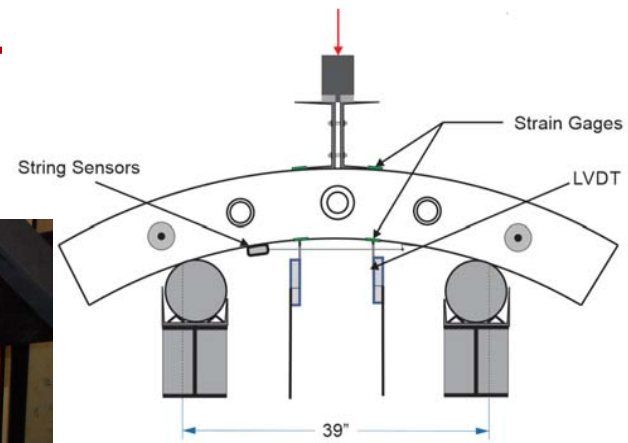


(d)

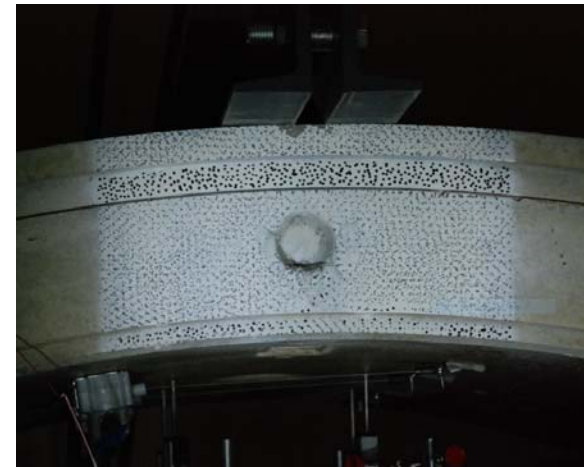
Testing and Instrumentation



3-Point bend test setup



DIC

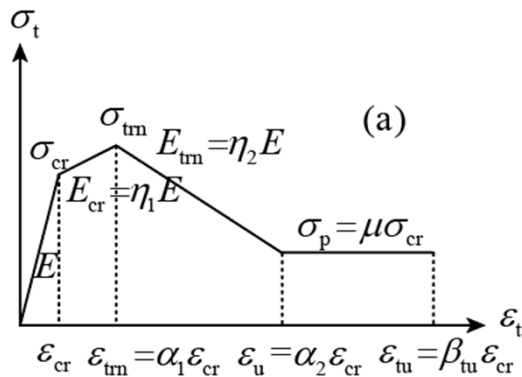
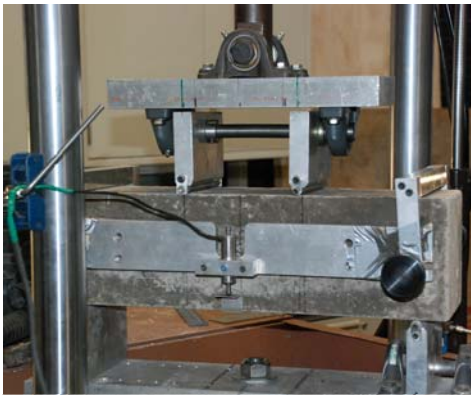


Load Deflection Response, SFRC

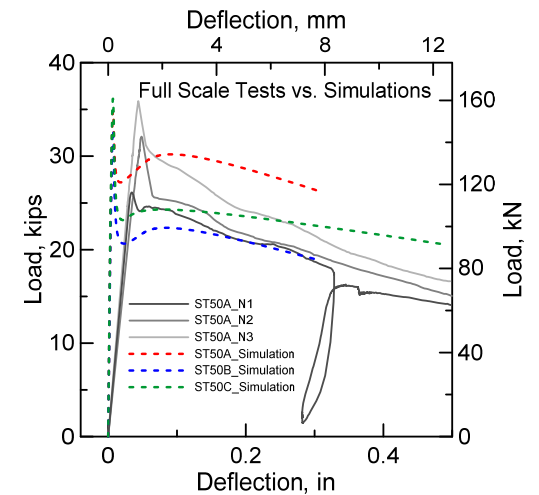
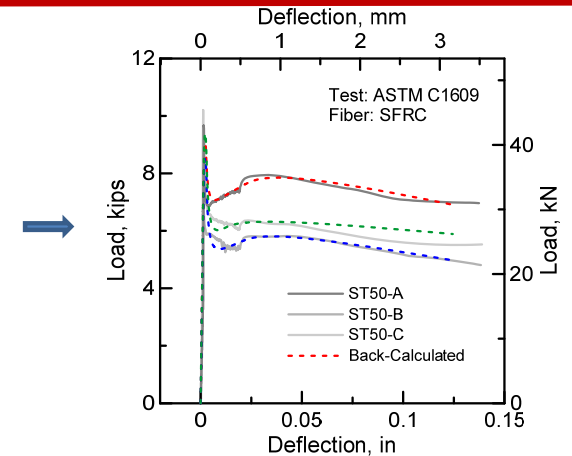
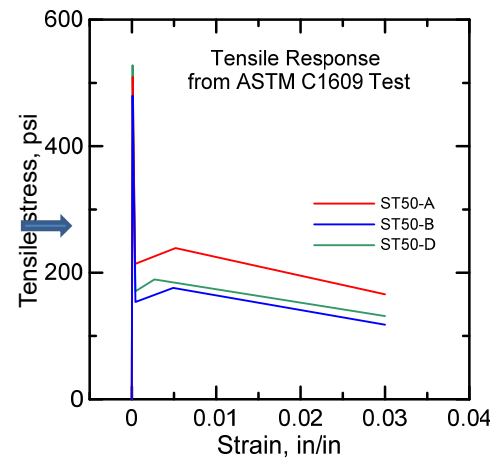
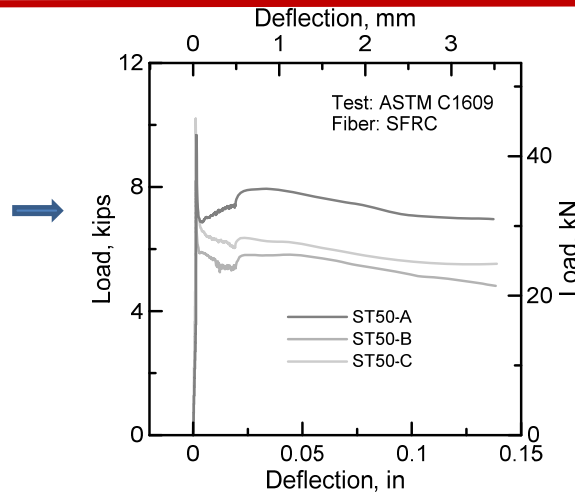


Cracked SFRC segment

Quad-Linear Modeling for Simulating Tunnel segments based on Flexural ASTM C1609 Tests



Back-Calculation based on quad-Linear Tensile response model



Conclusions

- Serviceability limit state design procedures allow us to:
 - i. Conduct ASTM C1609 to develop a parameterized approach, only two intrinsic parameters and all others as dimensionless parameters with scalar coefficients
 - ii. Use piecewise linear models, so solutions obtained in closed form.
 - iii. Do not try to measure strength, only look at specific operating mechanisms of load-displacement response
 - iv. Obtain algorithms for strain distribution, stress distribution, location of neutral axis, internal forces and moment arms, strain softening, strain hardening, hybrid reinforcement, compression and tensile continuous rebars, FRP strengthening, flexure, shear stresses, compression, axial bending interaction diagrams, shear cracking criteria.

Conclusions

- Analytical closed form equations to predict the load-deflection response of UHPC based on bilinear moment-curvature relationship were developed
- The methods apply to all type of flexural members exhibiting deflection-softening and hardening behavior
- The effect of normalized moment and normalized curvature for hybrid reinforcement system was verified
- Accuracy of the proposed model was identified by comparing the simulated data with experimental data